

CITRATE TOXICITY	Citrate → liver → metabolise citrate to generate bicarbonate → metabolic alkalosis
HYPOTHERMIA	- Blood stored at 2-6°C → cold blood to patient → hypothermia triggered ↑ chances of bleeding - Inline warmer → prevents development of hypothermia
HYPERKALEMIA	- Stored blood has supernatant fluid which can have potassium upto 60 meq/L which can cause cardiac arrhythmia - Near expiry blood will have dextrose depletion and hence Na-K pumps fail - Volume depletion due to AKI
HYPOKALEMIA	- Seen hours after transfusion is completed - pH is inversely proportional to potassium - In case of alkalosis shift of potassium inside cells occur and hypokalemia seen
TETANY	- Citrate binds to calcium - Hypocalcemia
DILUTIONAL COAGULOPATHY	

- These complications more common if multiple units given or even with single unit if quality control is not maintained

FEBRILE NON-HEMOLYTIC TRANSFUSION REACTION (FNHTR)

- M/c complication** seen after Transfusion and Temperature rise by approx. 1°C
- Diagnosis of exclusion and seen after ruling out the following
 - Transfusion transmitted bacterial infections
 - Hemolytic reaction - Mismatch blood transfusion (Type II Hypersensitivity reaction)
- ↑ incidence seen in grand multipara and those who have received multiple transfusions
 - Cause: antibody to donor leucocytes
- Prevention: Pre storage **Deleucocytation** and premedication with PCM

ALLERGIC REACTIONS

- Urticaria, wheals and rash
- If **>2/3rd body** involved: it is called severe
- Stop transfusion temporarily and give Diphenhydramine
- Vitals monitoring every 5 minutes
- Anaphylaxis - **Basophil mediated** histamine release
 - Can cause death of the patient due to laryngeal edema
 - Systolic BP ↓↓, Cyanosis, Rhonchi
 - Compensatory tachycardia
 - T/t - **Adrenaline (1:1000)** I/M, I/V fluids
 - Rare event



HEMOLYTIC REACTION (TYPE II HYPERSENSITIVITY REACTION)

Mediated by Ig M class of antibodies

HEMOLYTIC REACTION

Acute - Occur immediately

Chronic - 5-14 days later

- Can occur due to clerical error (technician, doctors and nurse)
 - Failure of correct product identification and patient identification
 - Error in blood product grouping and labelling
- Clinical features
 - Fever with chills and rigors
 - Chest pain, Flank pain
 - Hemoglobinuria (cola coloured urine)
 - Disseminated Intravascular Coagulation (DIC)
 - Bleeding from I/V cannula site
 - Tachycardia, hypotension, ↑ respiratory rate, ↓ SPO2
 - Shock, Acute kidney injury
- Work up
 - LDH: ↑
 - Bilirubin: unconjugated ↑ (intravascular hemolysis)
 - Haptoglobin: Absent as it binds to free hemoglobin
 - Direct coombs test - positive → demonstrates presence of bound antibody
 - Indirect coombs test - free antibody detection

T/t

Stop transfusion and send unit to blood bank, keep I/V line patent

I/V Fluid + **Furosemide** (to increase GFR) and soda bicarbonate - Forced Alkaline diuresis

Input output monitoring and watch on potassium and uremia, Pressor support

If DIC features, then give Platelet Concentrate and FFP

- Delayed reaction: 5-14 days later
 - Jaundice and pallor developing one week later after transfusion

TRANSFUSION ASSOCIATED GRAFT VS HOST DISEASES (TA-GVHD)

- Extremely rare
- Occurs due to Donor TB lymphocytes causing damage to recipient's organs

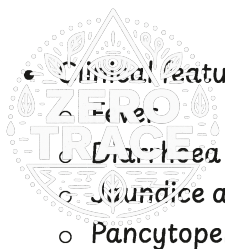
RISK FACTORS

Immunocompromised patient

Autologous bone marrow transplantation

Patient on **Antithymocyte globulin**

Blood transfusion from relative



Clinical features

- Diarrhoea (GIT)
- Jaundice and pruritus (Liver)
- Pancytopenia (Bone marrow)
- Death can occur within 3 weeks
- Prevention by using irradiated blood (most radiosensitive blood cell - Lymphocytes, dose of radiation used - 25Gy)

Important Information

- Most radioresistant blood cell - platelets

TRANSFUSION -RELATED ACUTE LUNG INJURY (TRALI)

- <6 hours of transfusion
- Incidence: more with fresh frozen plasma
- Can occur upto 72 hours
- Cause - transfusion of donor plasma containing
 - Anti-HNA antibody (human neutrophil antigen)
 - Anti HLA II antibody
 - Anti- HLA I antibody
- These antibodies (anti-HNA antibody) cause **leukocyte aggregation** in pulmonary circulation and release of cytokines that damage pulmonary endothelial cells → non cardiogenic pulmonary edema
- Prevention of TRALI
 - Transfusion of blood products from only male donor / nulliparous woman
 - Parous woman with absent anti HNA (human neutrophil antibody) antibodies
 - Pre storage Deleucocytation



DIAGNOSTIC CRITERIA

PaO₂/ FiO₂ <300 (normal ≥500)

Non cardiogenic pulmonary edema due to leaky endothelium

CXR- extensive B/L infiltrates

Important Information

- In cardiogenic pulmonary edema - BNP ↑
- In non-cardiogenic pulmonary edema BNP levels normal / slightly elevated
- T/t - **Supplemental oxygen** support and quarantine that unit of blood from that particular donor

TRANSFUSION ASSOCIATED CIRCULATORY OVERLOAD (TACO)

- Risk factors
 - Old age, patient with CAD, Renal failure
 - Administration of large volume blood components



- Normal transfusion time ~4 hours/unit
- Main cause of death after blood transfusion

CASE SCENARIO

00:53:08

Q. 50-year-old T2DM with CKD G4 A2 is scheduled for hemodialysis today. Pre dialysis Hb is 6 gm%. So 2 units of whole blood were transfused in last 5 hours. While second unit is transfused patient has severe SOB and drop in SaO₂. On examination SBP is elevated, pulses are bounding and B/L crepitations are heard over both lung fields. BNP levels are 5X elevated. Diagnosis?

DIAGNOSIS

- Cardiogenic pulmonary edema (TACO)
- BNP done

	TRALI	TACO
BP	N/↓	↑
JVP	Normal	↑
BNP Levels	n/↑	BNP >1000 pg / ml / rise >1.5x baseline
MANAGEMENT	Stop transfusion, Supplemental oxygen	Stop transfusion, I/V Furosemide

HYPERKALEMIA AND HYPOKALEMIA

- Stored blood being transfused, near expiry date blood (nonfunctioning Na-K pump)
- Post transfusion when metabolic alkalosis occurs due to citrate excess
- Chances of Hyperkalemia > hypokalemia

IRON OVERLOAD

- Occurs in people with Recurrent blood transfusion (thalassemia)
- 1 unit RBC = 200-250 mg iron
- Start Deferoxamine when serum ferritin >1000 ng/ml or cardiac MRI/ liver MRI report
- Leads to Bronze diabetes if not intervened - bronzing of skin, liver damage, type I DM, joint damage, heart damage, lung damage, restrictive cardiomyopathy

HYPOTENSIVE REACTION

- During transfusion, SBP falls >30 mmHg and when transfusion is stopped then SBP will normalise
- Risk factor
 - Patient on ACEI given blood transfusion
 - Leukocyte filters can activate Kallikrein-Kinin system causing excess bradykinin → hypotension
- Management - delay transfusion and control hypertension with alternate anti-hypertensive medication or hold ACEI 24 hours before transfusion if possible

INFECTIOUS ADVERSE REACTIONS

• Alert M/c

- Highest risk with platelet transfusion as it is stored at room temperature (20-24°C)
- Gram negative organism for transfusion transmitted bacterial infection - *Yersinia enterocolitica*, *Pseudomonas*, *Acinetobacter*
- Hepatitis B transmission risk: <0.5 per 10⁶ blood products (use of NAAT)
- Hepatitis C transmission <0.1-1% per 10⁶ blood products (use NAAT)
- Rarely transmitted after blood transfusion: **hepatitis A**

MCQ

Q. A 30-year-old multigravida develops fever and breathing difficulty while receiving 2nd unit of whole blood for management of postpartum hemorrhage. On examination pulse is 120/min, BP 120/90 mmHg, JVP is normal with B/L crepitation in B/L lung fields. CXR is shown. What is likely diagnosis?

- TACO
- TRALI
- Haemolytic reaction
- Allergic reaction

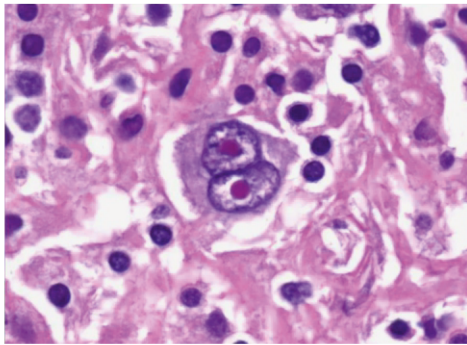
Ans. (b)



BURKITT'S & NON-HODGKIN'S LYMPHOMA

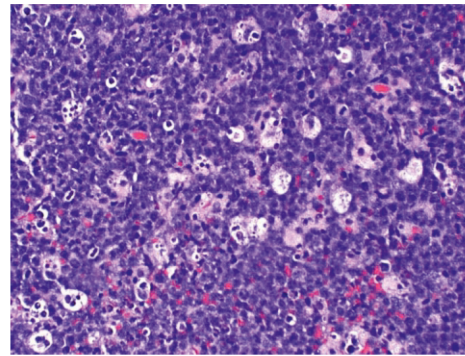
Hodgkins lymphoma

- HPE: **Reed Sternberg cells** → "Owl-eye" appearance



Burkitt lymphoma (NHL)

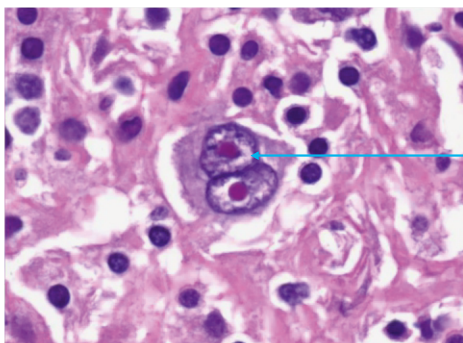
- HPE: "**Starry sky**" appearance
- ↑ Risk of **tumor lysis syndrome**



HODGKIN'S LYMPHOMA

00:00:48

- Cure rate: > 85%
 - Etiology:
 - EBV
 - HIV
 - Not included in AIDS- defining cancers
 - IOC: Excisional lymph node biopsy
- Lymph nodes show **polyclonal inflammatory infiltrate + Reed-Sternberg cells** (Owl eye appearance)



Reed Sternberg cell

Important Information

AIDS-Defining Cancers

- Kaposi sarcoma (aka KSHV)
- NHL: Primary CNS lymphoma
- Invasive Cervical Cancer

Important Information

- **Owl eye appearance** in **urine cytology** of a **post-kidney transplant** patient with graft failure → suggests **CMV infection**

- Genetic aberration → Amplification of **chromosome 9p24.1**
- **Mature B-cell tumor**

CLINICAL FEATURES

- **Bimodal** age distribution → peaks at ~20 yrs and ~80 yrs
- Progressive, painless **cervical lymphadenopathy** with **discrete, rubbery** masses



- Mediastinal lymphadenopathy causing:
 - Tracheal compression → SOB at rest, stridor
 - SVC compression → SVC syndrome
 - **M/c cause of SVC syndrome : small cell lung cancer**
 - **Thoracic duct compression** → Chylous ascites
- Porta hepatis lymphadenopathy → Causes obstructive jaundice / pruritus
- Retroperitoneal Lymphadenopathy causing **hydronephrosis**
- Category B Symptoms
 - Weight loss **>10%** in 6 months
 - Drenching night sweats
 - Loss of appetite
- **Pel-Ebstein fever**
 - Intermittent high-grade fever (Not considered reliable by hematologists)
- Paraneoplastic manifestations of HL
 - Nephrotic syndrome
 - Cerebellar degeneration
 - Guillain-Barré syndrome (GBS)
 - Hypercalcemia
 - Leading cause of hospitalization → **Squamous cell lung cancer** (produce PTH-rp)
 - Immune-mediated anemia, thrombocytopenia → Warm antibodies of IgG variety

Important Information

- Involuntary weight loss → >5% body weight or >10 pounds over 6 months

Important Information

- GBS can be associated with:
 - Campylobacter jejuni
 - Zika virus
 - Hepatitis E virus
 - Hodgkin Lymphoma

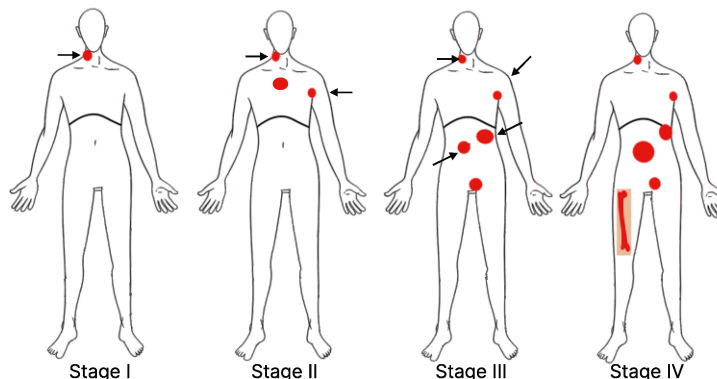
WORKUP

00:13:18

- CBC
- LDH ↑
- Imaging: for staging
 - PET-CT
 - CT chest / CT abdomen
- **IOC**: Excisional lymph node biopsy
- Pulmonary Function Test
- Echocardiography
- Infection panel

MODIFIED ANN ARBOR CLASSIFICATION

Stage I	• Single LN group or lymphoid tissues (spleen, thymus)
Stage II	• ≥2 LN groups on the same side of the diaphragm
Stage III	• ≥2 LN groups on both sides of the diaphragm
Stage IV	• Extranodal extension (e.g., liver, bone marrow)



- A: No category B symptoms
- B: Category B symptoms
- E: Extra lymphatic site, except liver & bone marrow (e.g., mediastinal LN + lung involvement)

WHO MODIFIED REAL CLASSIFICATION

00:18:48

Classical Hodgkin Lymphoma: CD15⁺, CD30⁺

- Nodular sclerosis:
 - M/c variety overall
 - HPE: **Lacunar cells**
- Mixed cellularity
 - M/c in India
 - HPE: Classical Reed-Sternberg (RS) cells
 - Associated with HIV
- Lymphocyte-rich
- Lymphocyte-depleted

Non-Classical or Atypical HL: CD15⁻, CD30⁻

- Nodular Lymphocyte-Predominant Type
 - **L & H cells / Popcorn cells**
 - CD15/30: **Negative**
 - CD19, CD20, CD45 & CD79a : **Positive**
 - May transform into **diffuse large B-cell lymphoma (DLBCL)**
 - If progression → RT and R-CHOP

TREATMENT OF HODGKIN'S LYMPHOMA

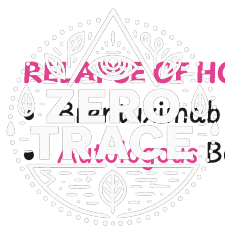
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- Localized: ABVD ± Radiotherapy
- Advanced: 6 cycles of ABVD
 - **A**driamycin (Doxorubicin)
 - **B**leomycin
 - **V**inblastine
 - **D**acarbazine

Important Information

- Hodgkin's lymphoma spreads via lymphatics
- Follows contiguous spread

Drug	Key Toxicity
Bleomycin	Pulmonary fibrosis (monitor with PFTs)
Doxorubicin/ Adriamycin	Cardiotoxicity (dose-limited)
Vinblastine	Neurotoxicity, constipation
Dacarbazine	Myelosuppression, nausea



RELAPSE OF HODGKIN'S LYMPHOMA

Relapse of Hodgkin's Lymphoma

- Autologous Bone Marrow Transplantation (ABMT)

INTERNATIONAL PROGNOSTIC INDEX (IPI)

- Age, Gender, LDH
- Hemoglobin, Albumin, TLC, Lymphocyte count

NON-HODGKIN'S LYMPHOMA

ETIOLOGY

- Epstein-Barr Virus (EBV)
- HIV
- HTLV-I: Adult T-cell lymphoma
- HHV-8 / KSHV: Primary effusion lymphoma
- HCV: Lymphoplasmacytic lymphoma
- H. pylori: Gastric MALToma
 - Chromosomal swap: t (11;18)
- Chromosomal swaps
 - t (14;18): M/c
 - t (11;14): Mantle cell lymphoma

BURKITT LYMPHOMA

- Histopathology: Starry sky appearance
- Rapid tumor doubling time & ↑ risk of tumor lysis syndrome (TLS)
- Chromosomal Translocations:
 - t (8;14): M/c
 - t (2;8)
 - t (8;22)
- Overexpression of c-myc d/t t(8;14)(q24;q32) → inhibits apoptosis
- Presentation: African child with jaw swelling

Important Information

Autologous BMT is used in:

- Multiple myeloma
- Relapsed Hodgkin's lymphoma

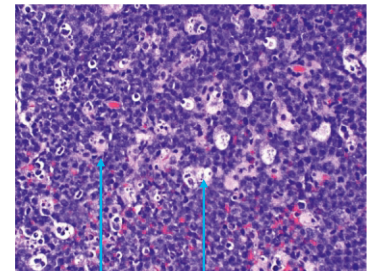
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Important Information

Hematological tumors caused by EBV

- Burkitt Lymphoma
- Diffuse Large B-cell Lymphoma
- Hodgkin's Lymphoma
- Extranodal NK / T-cell lymphoma

00:32:11



Night sky Star



Typical presentation in Burkitt's

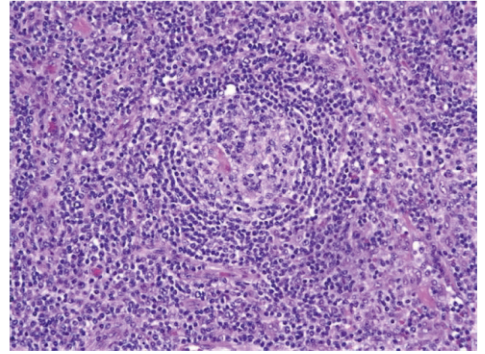
Extra Mile

Castleman Syndrome

- Etiology: KSHV (HHV-8)
- HPE: "Onion-ring" appearance on LN biopsy

HIV-Associated Conditions:

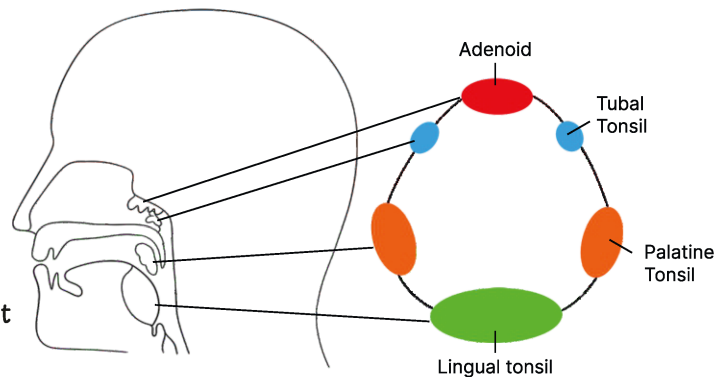
- Kaposi Sarcoma (KS)
- Primary Effusion Lymphoma
- Castleman Syndrome



"Onion-ring" appearance

CLINICAL FEATURES

- Usually seen in adults
- Abdominal lymph node enlargement
 - Intestinal obstruction
 - Ascites
 - Hydronephrosis (due to ureteral compression)
 - Testicular lump
- Meningeal Spread
 - ↑ Intracranial pressure (ICP)
 - Meningismus
- Category B symptoms
 - **Drenching night sweats**
 - Weight loss
- Bone marrow involvement
 - **Cytopenia**
 - Normocytic Normochromic Anemia: ↓ RBC count
 - ↓ Platelets
- Spinal cord compression, Cranial nerve palsy
- **Waldeyer's ring involvement** (tonsillar involvement: Snoring↑)
- Erythroderma (Sezary syndrome / CTCL)



WORK UP

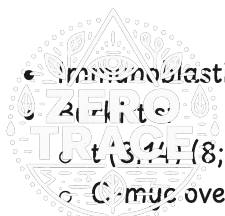
- CBC
- LDH ↑
- PET-CT: Staging
- MRI spine
- FISH: To identify chromosomal translocations
- **IOC**: Excisional lymph node biopsy

WHO MODIFIED REAL CLASSIFICATION

B CELL TUMORS

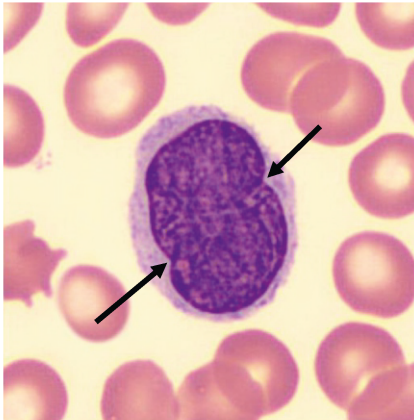
- DLBCL
 - M/c
 - Associated with HIV, EBV

00:42:27



- Immunoblastic lymphoma: Associated with HIV
 - t(3;14)(8;22) (2;8)
 - C-myc overexpression leading to inhibition of apoptosis
- MALToma: d/t H. Pylori
- Lymphoplasmacytic lymphoma: HCV

T CELL NHL

MYCOSIS FUNGOIDES	• Presents as progressive eczema • Skin biopsy shows Pautrier microabscess formation	
SEZARY SYNDROME	• Dissemination of cutaneous T-cell tumor ◦ Erythroderma ◦ Hepatosplenomegaly ◦ Cytopenias: Hb↓, Platelet ↓ ◦ Malignant CD4⁺ cells with cerebriform appearance	 Sezary Cell

INTERNATIONAL PROGNOSTIC INDEX

- Age
- LDH
- Stage
- Extranodal site involvement
- ECOG performance status

TREATMENT

- R CHOP
 - **R**ituximab (Anti CD20 molecule)
 - **C**yclophosphamide (Alkylating agent)
 - **H**ydroxy daunomycin
 - **O**ncovin / **V**incristine
 - **P**rednisolone



Q. A 45-year-old male presents with painless cervical lymphadenopathy with no extranodal involvement. Flow cytometry reveals cells positive for CD20, CD19, and CD45 but negative for CD30 and CD15. Diagnosis?

- a. Follicular Lymphoma
- b. Diffuse Large B-cell Lymphoma
- c. Classical Hodgkin's Lymphoma
- d. Nodular Lymphocyte-Predominant Hodgkin's Lymphoma

Ans: (d)

BETA THALASSEMIA/ COOLEY'S ANEMIA

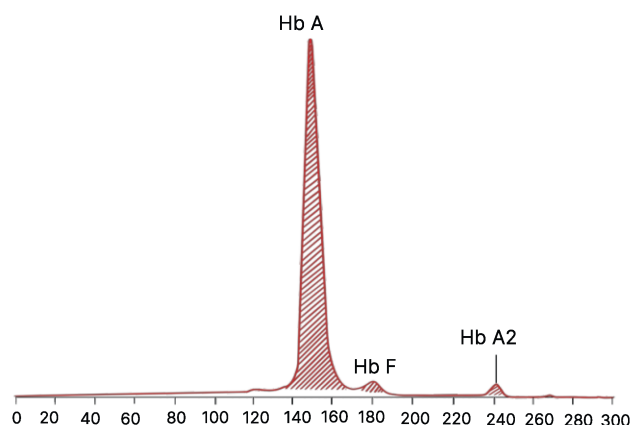
- Inheritance: Autosomal recessive
- Gene Defect: **HBB gene** on **Chromosome 11**
- Mutation Type: Single nucleotide change > nonsense mutation
- Feature: **Chipmunk facies**



Chipmunk facies

HEMOGLOBIN ELECTROPHORESIS

	AT BIRTH	6 MONTHS WHEN ADULT VALUES ARE OBTAINED
Hb F	95%	2%
Hb A ($\alpha_2\beta_2$)	2%	95%
Hb A2	3%	3%



BASIC SCIENCE CONCEPTS

- Earliest embryonic hemoglobin - Gower 1 > Gower 2 > Portland Hb
- Hematopoiesis starts at **2 weeks** in Yolk sac
- At **14 weeks** it shifts to liver and spleen and HbF is formed
- At **38 weeks** it begins in bone marrow and HbA appears/formed
 - Bones involved - Skull, Ribs, Vertebrae, Long bones, Flat bones

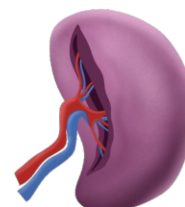
BETA THALASSEMIA MAJOR

- Normal RBCs contain hemoglobin with 2 alpha and 2 beta chains
- In β -thalassemia major, beta chains are absent, only alpha chains are produced
- Excess alpha chains aggregate to form α_4 tetramers
 - ~99% destroyed in bone marrow → **ineffective erythropoiesis**
 - Remaining RBCs destroyed in spleen → **extravascular hemolysis**
- Extramedullary hematopoiesis: Liver and spleen starts producing RBCs
 - This leads to ↑ size of liver and spleen - Spleno-hepatomegaly
- <1year:
 - Severe anemia → CHF
 - Spleno-hepatomegaly
 - Chipmunk facies



2 α chains
No β chain

Ineffective Erythropoiesis



Extravascular Hemolysis

00:07:01

IMPORTANT FEATURES

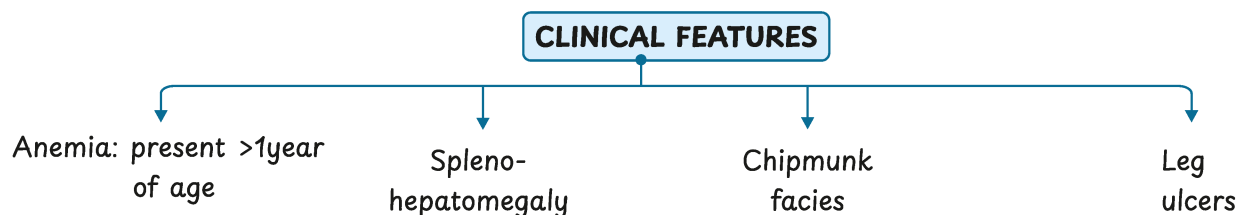
	• Transfusion dependency, Target cells on peripheral smear
H	• Spleno-hepatomegaly
A	• Anemia (CHF)
L	• Low MCV, MCH, Leg ulcers
A	• Asian/Mediterranean origin
S	• Skull bossing
E	• Extramedullary hematopoiesis, Elevated HbA ₂ , HbF
M	• Marrow expansion
I	• Iron Overload - need for chelation
A	• Abnormal electrophoresis

- **IOC** - HBB gene sequencing > **HPLC** > **Electrophoresis**

BETA THALASSEMIA INTERMEDIA

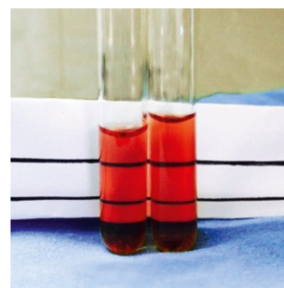
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- 50% β -chain supply present
- Clinical features: lesser severity of manifestation

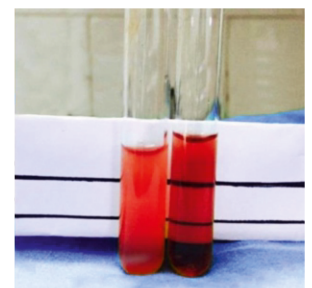


BETA THALASSEMIA TRAIT (MINOR)

- **Asymptomatic/mild** anemia with low MCH, RDW $\uparrow$
- Screening test: **NESTROFT** (Naked Eye Single Tube Red Cell Osmotic Fragility Test)
 - Unlysed RBC in hypotonic solution will make sample hazy
 - RBCs immersed in hypotonic solution (0.36% saline) $\rightarrow$ Osmotic fragility $\downarrow$
- Additional test: **HbA₂ > 3.0% (WHO)**



NESTROFT Test - Negative



NESTROFT Test - Positive



PRENATAL DIAGNOSIS OF THALASSEMIA

Chorionic villus sampling (CVS) at 10-12 weeks of gestation

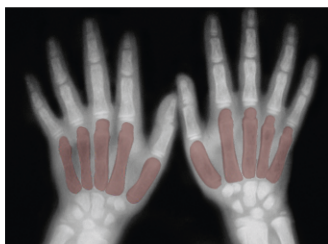
Amniocentesis at 16 weeks

Cordocentesis at 18-20 weeks

WORKUP

00:24:42

- CBC - ↓ Hemoglobin (Hb), RDW >14%
- Peripheral smear -
 - Target cells
 - Microcytic hypochromic Anemia
- Reticulocyte count - ↑
- NESTROFT test - Screening test
- **IOC** -
 - HBB gene sequencing > HPLC (High-Performance Liquid Chromatography)
- X-Ray Hands and Skull



Squaring of Metacarpals

X-Ray Hands

- Squaring of metacarpals
- Appears early

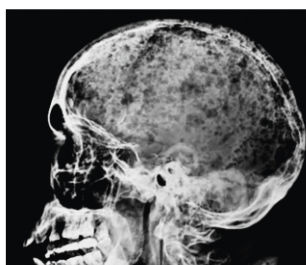


Hair-on-end appearance

X-Ray Skull

- Widening of diploic spaces → bone trabeculations appear more prominent → **Hair on end/Crew cut** appearance
- Late manifestation

Extra mile IBQs on X-Ray Skull



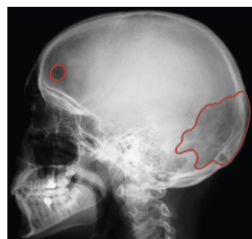
Old man with Raindrop lesions in skull, Diagnosis?

Ans: **Multiple Myeloma**



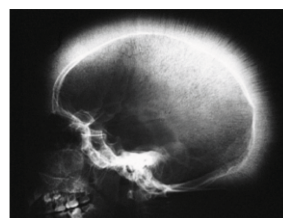
Old man with Hearing loss, change in shape of skull and SAP elevation. X-ray skull is given. Diagnosis?

Ans: **Paget's Disease**



Geographic Skull

Manifestation of **Histiocytosis X**



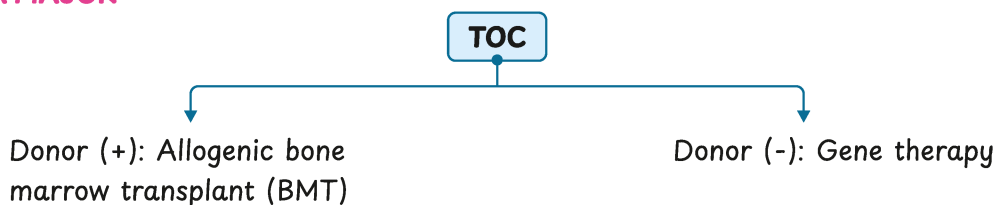
Hair-on-end appearance

Seen with **SCA and Thalassemia**

T/t

00:29:57

THALASSEMIA MAJOR



- For survival: Regular packed RBC transfusions (Leucodepleted blood units)
 - Frequency: **Every** 2–4 weeks
 - Lifespan of transfused RBCs: **60–80** days
 - 1 Unit of PRBC delivers 200–250 mg **iron**, leading to secondary hemochromatosis, Bronze diabetes
- Excess iron ↑ risk of sepsis d/t **yersinia enterocolitica**
- **Luspatercept**: ↓ transfusion dependency

HOW TO EVALUATE FOR SECONDARY HEMOCHROMATOSIS AND DECIDE NEED FOR CHELATION?

- Cardiac MRI T2 relaxation time < 20ms
- Liver MRI
- Iron profile: Serum ferritin > 1000 ng/mL
- MOA of Luspatercept - enhances late-stage erythropoiesis

Important Information

- Acute iron toxicity - Serum iron > 500 µg/dl indicate serious iron toxicity

PARAMETER	CUT-OFF	ACTION
Serum Ferritin	> 1000 ng/mL (chronic transfusion)	Start iron chelation
After Bone Marrow Transplant (BMT)	Ferritin > 500–1000 ng/mL + evidence of iron overload	Start chelation cautiously
Transfusion Volume	After 10–20 transfusions OR 120 mL/kg RBCs	Start chelation even if ferritin borderline



Liver MRI (LIC = Liver Iron Concentration)	LIC > 3 mg Fe/g dry weight	Start iron chelation
Cardiac MRI (T2)	T2 < 20 ms	Start or intensify iron chelation urgently

ALPHA THALASSEMIA

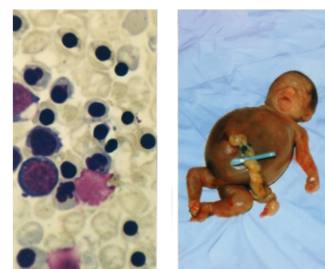
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- Autosomal recessive
- Genes involved: **HBA1**, **HBA2** genes
- Chromosome: 16
- HbA has 2 alpha and 2 beta chains
 - 2 Beta genes involved for synthesis of 2 beta chains
 - 4 alpha genes involved in production of 2 alpha genes

GENES	α CHAIN PRODUCTION	
αα/αα	Normal	
-α/αα	Only 75% production	• α ⁺ Thalassemia
--/αα	Only 50%	• α ⁰ Thalassemia
--/-α	Only 25%	• K/α HbH Disease • Formation of beta 4 tetramers - ◦ High oxygen affinity ◦ Unstable Hb • Peripheral smear - Golf ball inclusion seen in brilliant cresyl blue
--/--	0%	• Gamma 4 Tetramers (Hb Barts)

Hb BART'S

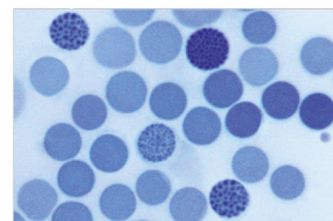
- Defect: γ₄ Tetramers
- High oxygen affinity, ↓ delivery to tissues
→ ↓pO₂ → CHF → Edema
- **Non-immunogenic** Hydrops fetalis

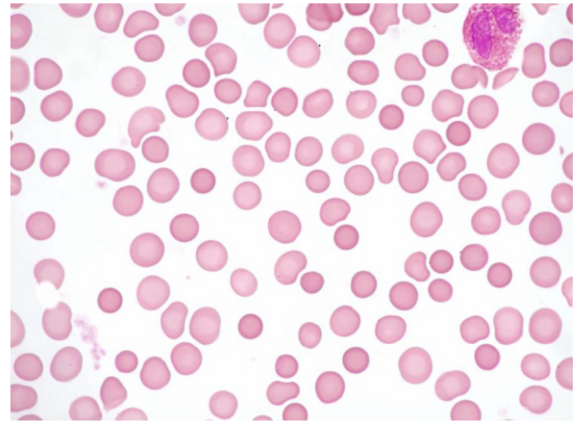
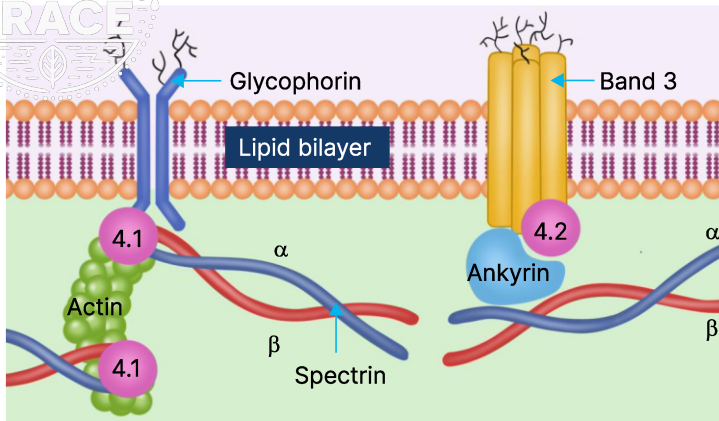


Q: A child visiting India for medical T/t presents with progressive anemia and hepatosplenomegaly. A peripheral blood smear reveals the given findings using a special stain. What is the most likely diagnosis?

- HbH disease
- G6PD deficiency
- Zika virus infection
- Bernard Soulier syndrome

Ans. (a)





- RBC Membrane contains Spectrin, Ankyrin, Protein 4.2, Band 3
- Defect in ankyrin - **ANK1 gene, chromosome 8**
- Development of rigid RBCs (spherocytes)
- Spherocytes on peripheral smear - lack central pallor
- RBC shape distorted → ↓ deformability → RBCs get trapped in spleen → extravascular hemolysis

CLINICAL FEATURES	• Pathological jaundice → require phototherapy • Pallor → Hb↓, evidence of Extravascular hemolysis • Splenomegaly • Gallstones → Pigmented (Black) • Non-healing ulcer → Medial malleolus/ leg ulcers • Aplastic crisis → Human parvovirus B19
WORK-UP	• Hemoglobin ↓ • Peripheral smear → Spherocytes • Direct Coombs Test → Negative • Reticulocyte count ↑ • MCHC > 34, RDW > 14, ↓MCV - highly suggestive • LDH ↑ • Bilirubin ↑ (↑ unconjugated) • Haptoglobin ↓ (mop for free Hb) • NESTROFT test - Osmotic fragility ↑ ◦ In Thalassemia - Osmotic Fragility ↓
SPECIAL TESTS	• Osmotic gradient ektacytometry: More sensitive than NESTROFT, but limited availability • Eosin-5-maleimide by flow cytometry: Most sensitive and specific test • SDS-PAGE for RBC membrane protein analysis • Acidified glycerol test



	• Positive family history of anemia with jaundice and - ○ Reticulocyte ↑, LDH ↑, Serum Bilirubin ↑ (unconjugated), Haptoglobin ↓ ○ Peripheral smear shows Spherocytes, NESTROFT ↑↑, Coombs Negative ○ MCHC ≥ 34 or right shift of nomogram
T/t	• Severe HS - Elective splenectomy at ~4 months • Moderate & Mild HS - Elective splenectomy ○ Delayed, done after puberty • All cases - Folic acid supplementation • Vaccines before splenectomy - ○ Hib, N. meningitidis, Pneumococcus • Post-splenectomy peripheral smear findings- ○ Cabot rings, Howell-Jolly bodies

MCQ

- Q. A 5-year-old boy presents with fatigue and intermittent jaundice. His blood film shows spherocytes, and his parents report a family history of similar symptoms. Laboratory tests reveal elevated indirect bilirubin and reticulocytosis. Which test is best for diagnosis?
- Osmotic gradient ektacytometry
 - NESTROFT
 - Eosin 5 Maleimide flow cytometry
 - Spectrin assay with SDS PAGE

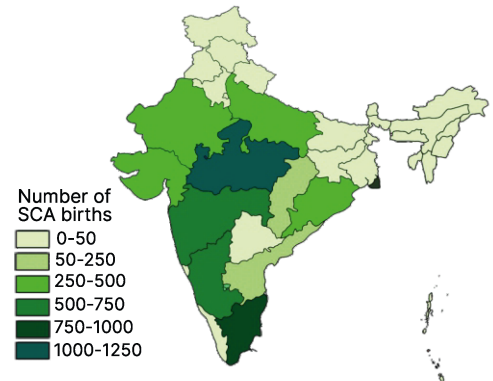
Ans. (c)



12. SICKLE CELL DISEASE, G6PD DEFICIENCY AND OTHER HEMOLYTIC ANEMIAS

SICKLE CELL ANEMIA

- **AR, Point mutation in HBB gene on chromosome 11**
- Beta 7 position, Glutamate replaced by Valine
- This positional swapping will result in formation of HbS
- HbS polymerization → RBCs become rigid and sticky
 - Rigid RBCs → trapped in splenic sinusoids → extravascular hemolysis
 - Sticky RBCs → microinfarctions → autosplenectomy



CLINICAL FEATURES

- Child from tribal belt/Afro-American child with
 - Fever
 - Bone Pain
 - Avascular necrosis of bone
 - **Hand-foot syndrome** / Dactylitis
 - Osteomyelitis risk (Salmonella, Staph aureus)
 - **Fish vertebra**
 - Bone within a bone, opioid addiction
 - Acute Abdomen
 - Repeated hospitalizations
 - Recurrent Splenic infarctions
 - Autosplenectomy occurs by 3 years and makes patient prone to infections with encapsulated organisms
 - Pneumococcus
 - Meningococcus
 - Haemophilus influenzae
 - Microinfarction at coronary perforators:
 - Chest pain
 - Dilated cardiomyopathy
 - Pulmonary Artery: Vaso-occlusion of small branches
 - **Acute chest syndrome**: Fever + hypoxia + chest pain
 - Pulmonary artery hypertension
 - Auscultatory finding: **Loud P₂**
 - Kidney: Papillary necrosis: Hematuria, obstructive uropathy d/t sloughed tissue and UTI, CKD
 - Causes of Papillary Necrosis:
 - **P**yelonephritis
 - **O**bstructive uropathy
 - **S**ickle Cell Anemia